Computer Vision-Based Interaction Using Webcam for Digital Games: An Experimental Study

[OMITTED FOR BLIND REVIEW]

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Abstract. ***Introduction:*** *Camera-based interaction expands how players control digital games by using body movement as an input modality. Low-cost webcams and open-source computer vision libraries make this approach accessible, although its effectiveness depends on reliable motion detection, low latency, and clear interaction mapping.* ***Objective:*** *This study develops and evaluates a webcam-based interaction prototype for digital games, focusing on the feasibility of motion-driven commands for simple gameplay actions.* ***Methodology:*** *The prototype uses OpenCV to capture video from a conventional webcam, preprocess frames, detect player movement, and map motion events to game commands, following common vision-based interaction procedures [Bradski 2000, Wachs et al. 2011].* ***Expected Results:*** *The study is expected to demonstrate the feasibility of webcam-based motion detection for accessible and engaging interaction in short gameplay scenarios, while identifying technical limitations related to lighting, player distance, and command precision, as reported in prior computer vision studies [Szeliski 2022, Piccardi 2004].*
Keywords *Computer Vision; Game Interaction; Human-Computer Interaction; OpenCV; Motion Detection*

1. Introduction

Digital games have traditionally relied on keyboards, mice, gamepads, and touchscreens as their main input devices. Although these interfaces are effective, they may limit interaction possibilities in contexts where body movement, accessibility, or low-cost experimentation are relevant. Computer vision offers an alternative by allowing a camera to interpret player actions and transform them into commands, a direction aligned with natural user interfaces and gesture-based human-computer interaction [Dix et al. 2004, Wachs et al. 2011].

Webcams are widely available and inexpensive, which makes them suitable for experimental game interaction systems. Combined with libraries such as OpenCV, they can be used to detect motion, segment regions of interest, and trigger actions without requiring specialized sensors [Bradski 2000]. However, webcam-based interaction remains sensitive to lighting conditions, background variation, camera position, and processing latency, which are recurring constraints in video analysis and real-time interactive systems [Szeliski 2022, Card et al. 1983].

This paper presents an experimental study on computer vision-based interaction using a webcam for digital games. The work examines how motion detection can be implemented and evaluated as an input mechanism for simple gameplay actions. The main contribution is a concise method for developing and assessing a prototype that connects real-time video processing to game interaction while considering the practical trade-off between recognition accuracy and response latency.

2. Related Work

Human-computer interaction research has explored natural user interfaces that reduce dependence on traditional controllers. In games, motion-based interaction can increase player engagement by connecting physical movement to in-game actions, provided that the mapping between action and feedback remains understandable to users [Dix et al. 2004, Norman 2013]. Commercial systems have demonstrated the potential of camera-based input, but such solutions often depend on dedicated hardware or proprietary platforms.

Computer vision techniques provide a flexible foundation for alternative interaction models. Frame differencing, background subtraction, contour detection, and optical flow are commonly used to identify motion in video streams [Szeliski 2022, Lucas e Kanade 1981]. Background subtraction methods are especially relevant when the camera is static, although they require adaptation to illumination changes and dynamic scenes [Stauffer e Grimson 1999, Piccardi 2004]. OpenCV is frequently adopted in academic and prototyping contexts because it offers accessible implementations of these techniques and can be integrated with game engines or standalone applications [Bradski 2000].

For digital games, the main challenge is not only detecting movement but also translating it into meaningful, responsive, and consistent commands. Prior work on real-time pose recognition demonstrates the relevance of vision-based body interpretation for interactive systems [Shotton et al. 2011]. Gesture recognition surveys also indicate that robust interaction depends on segmentation quality, temporal consistency, and clear command vocabularies [Mitra e Acharya 2007, Wachs et al. 2011]. Interaction mappings must be simple enough to avoid false positives while still supporting enjoyable gameplay. Therefore, experimental evaluation is necessary to understand whether a webcam-based approach can provide acceptable responsiveness and usability under practical conditions.

3. Methodology

This study develops a prototype in which a conventional webcam captures the player's upper-body movement and converts selected motion patterns into game commands. The system architecture includes video acquisition, frame preprocessing, motion detection, command mapping, and gameplay feedback. OpenCV performs the image processing operations, supporting real-time prototyping through established computer vision routines [Bradski 2000], while the game layer receives discrete commands such as move, jump, select, or avoid.

The video processing pipeline begins by resizing frames and converting them to grayscale to reduce computational cost, a common preprocessing step in image analysis pipelines [Szeliski 2022]. Basic filtering reduces noise, followed by motion detection through frame differencing or background subtraction, following established computer vision procedures for static-camera video analysis [Stauffer e Grimson 1999, Piccardi 2004]. Detected contours are analyzed according to area, position, and variation over time, using shape and region information to distinguish intentional movement from minor visual changes, following contour extraction principles used in image border analysis [Suzuki e Abe 1985]. When a movement satisfies predefined thresholds, the prototype generates the corresponding game action.

The pipeline is organized as a real-time loop with four stages: frame capture, visual preprocessing, motion interpretation, and command dispatch. Each frame is timestamped before processing so that end-to-end latency can be estimated from acquisition to command emission, since response time directly affects perceived control in interactive systems [Card et al. 1983]. Thresholds for contour area and motion persistence are calibrated during a short preparation phase, reducing accidental activations caused by minor body movement or camera noise. This design favors simple and explainable motion rules instead of complex classification models, which is appropriate for a short experimental prototype and supports reproducibility across conventional webcams. Although full-body pose recognition can provide richer semantic information, prior real-time pose recognition research also highlights the computational and modeling demands of body interpretation [Shotton et al. 2011]; therefore, the proposed prototype uses motion cues as a lightweight alternative.

The experimental procedure involves controlled gameplay sessions using simple tasks that require repeated motion-based commands. Data collection focuses on recognition consistency, response time, observed errors, and participant perception. Quantitative measures include correctly detected commands, false activations, missed activations, and average interaction latency. These measures reflect core concerns in real-time vision systems, where recognition quality and processing time must be evaluated together [Szeliski 2022, Liu e Layland 1973]. Qualitative feedback is collected through short post-session questions about comfort, clarity, and perceived control.

To preserve reproducibility, the evaluation defines the camera distance, lighting setup, background condition, and sequence of interaction tasks. Participants perform repeated command trials under the same interaction script, allowing detected commands to be compared with expected commands. The analysis reports accuracy as the proportion of correctly recognized commands and treats false activations and missed activations as separate error categories. Latency is analyzed together with accuracy because stricter thresholds may reduce false positives while increasing response delay, whereas permissive thresholds may improve responsiveness but reduce precision.

The evaluation also records contextual observations, including player distance from the camera, body occlusion, and visible background motion. These observations help explain failures that may not be captured by aggregate measures alone. This method supports a focused assessment of feasibility without extending the scope beyond a short experimental study.

4. Expected Results

The prototype is expected to demonstrate that webcam-based motion detection can support simple digital game interactions using accessible hardware and open-source tools. The most promising results are anticipated in scenarios with stable lighting, limited background movement, and clearly separated player gestures, since image-based motion analysis is sensitive to visual noise, illumination changes, and scene variation [Szeliski 2022, Piccardi 2004].

The study is also expected to reveal constraints that affect practical adoption. Variations in illumination, camera angle, player distance, and background complexity may reduce detection accuracy, as observed in computer vision pipelines that depend on consistent visual evidence [Stauffer e Grimson 1999, Piccardi 2004]. Fast or subtle movements may cause missed commands, while excessive sensitivity may generate false activations. These results are expected to show the trade-off between accuracy and latency: more restrictive thresholds may improve precision but delay command recognition, whereas permissive thresholds may improve responsiveness at the cost of stability. Such latency constraints are relevant because delayed feedback can reduce the user's sense of control in interactive tasks [Card et al. 1983, Dix et al. 2004]. OpenCV enables rapid implementation of these parameter choices [Bradski 2000], but the interaction design must still balance computational cost and recognition reliability.

From a user experience perspective, the expected outcome is that players will perceive the interaction as engaging when commands are easy to understand and the system responds quickly. At the same time, fatigue and imprecise control may limit longer sessions. The comparison with pose-oriented approaches [Shotton et al. 2011] and gesture recognition literature [Mitra e Acharya 2007, Wachs et al. 2011] is expected to clarify when lightweight motion detection is sufficient and when richer body interpretation may be necessary. The results will therefore contribute to understanding where low-cost computer vision interaction is most appropriate in digital games.

5. Conclusion

This paper presents an experimental study of webcam-based interaction for digital games using computer vision techniques. The prototype uses OpenCV to detect player movement and map it to gameplay commands, emphasizing accessibility, low-cost hardware, and concise evaluation [Bradski 2000, Wachs et al. 2011].

The study is expected to clarify the feasibility and limitations of motion-based interaction through conventional webcams. By analyzing responsiveness, recognition consistency, and user perception, the work aims to support future improvements in computer vision interfaces for games. Further development may include more robust gesture recognition, adaptive thresholds, and evaluation with broader participant groups, drawing on established computer vision, background modeling, and body recognition methods [Szeliski 2022, Stauffer e Grimson 1999, Shotton et al. 2011].

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